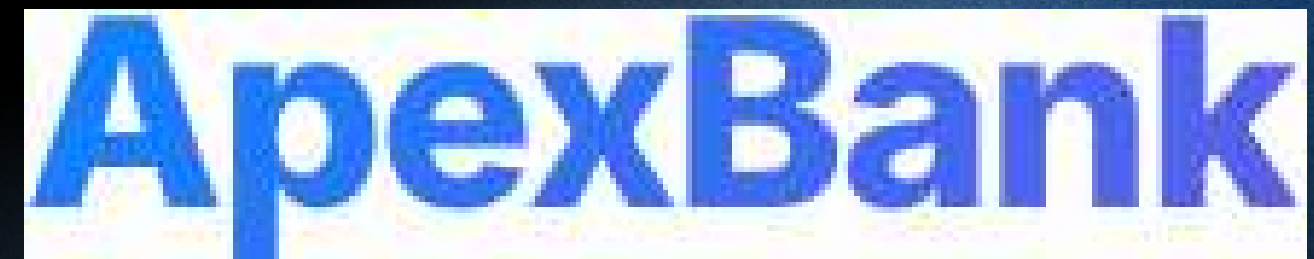


CAPSTONE PROJECT



ONLINE BANKING PORTAL

DEVELOPED BY

SUSHANK

OUTLINE

- Introduction
- Aim of the Project
- Objectives
- Functionalities of the Project
- System Requirements
- Technologies Used
- Project Modules
- User Characteristics

INTRODUCTION

- ApexBank is a modern, full-stack online banking application built to simulate real-world digital banking operations.
- Enables customers to open accounts, manage profiles, and transfer funds securely from anywhere.
- Replaces traditional branch-dependent banking with a fast, self-service digital platform.
- Designed using industry-standard practices: microservices architecture, secure authentication, and a modern responsive UI.
- Built with Spring Boot (backend) and Angular (frontend), following real banking system workflows.

AIM OF THE PROJECT

- To design and develop a secure, scalable online banking system that mirrors real-world banking operations.
- To provide customers a seamless digital experience for account management and fund transfers.
- To provide administrators a centralized dashboard to manage users, approvals, and transactions.
- To demonstrate practical implementation of microservices architecture in a real-world use case.

OBJECTIVES

- Implement secure user registration, login, and authentication using JWT.
- Allow new customers to apply for an account, subject to Admin approval.
- Provide a personalized dashboard showing account balance, UPI ID, and recent activity.
- Enable fund transfers through NEFT and UPI with real-time balance updates.
- Maintain a detailed, filterable account statement of all transactions.
- Build a dedicated Admin Portal for account approvals and user management.
- Apply microservices principles for scalability, fault isolation, and independent deployment.

FUNCTIONALITIES OF THE PROJECT

- User Registration & Login – Secure sign-up and authentication with account lockout after failed attempts.
- Account Opening – New account requests submitted by users, reviewed and approved by Admin.
- Dashboard – Snapshot of balance, UPI ID, and recent transactions.
- Account Statement – Date-range filtered transaction history.
- Profile Management – View and update personal and address details.
- Fund Transfer – Send money via NEFT (account number) or UPI (UPI ID / account number).
- Payee Management – Save and manage frequently used beneficiaries.
- Admin Portal – Approve/reject accounts, view all users, add money to accounts, and monitor all transactions system-wide.

SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Intel i3 or above
- RAM: 8 GB minimum
- Storage: 10 GB free disk space

Software Requirements

- Operating System: Windows 10/11, macOS, or Linux
- Java Development Kit (JDK) 17
- Node.js 18+ and Angular CLI 17
- MySQL Server 8.0
- IDE: Eclipse (backend), VS Code (frontend)
- Web Browser: Chrome, Edge etc

TECHNOLOGIES USED

Backend

- Spring Boot (Microservices Architecture)
- Spring Data JPA, JWT Authentication
- Netflix Eureka (Service Discovery), Spring Cloud Gateway (API Gateway)
- OpenFeign (Inter-service Communication)
- MySQL (Separate database per microservice)

Frontend

- Angular 17 (Standalone Components)
- Reactive Forms & Validators
- HTTP Interceptors & Route Guards
- Custom VisionOS-style Glassmorphism UI (CSS)

Tools

- Maven, Git, Postman

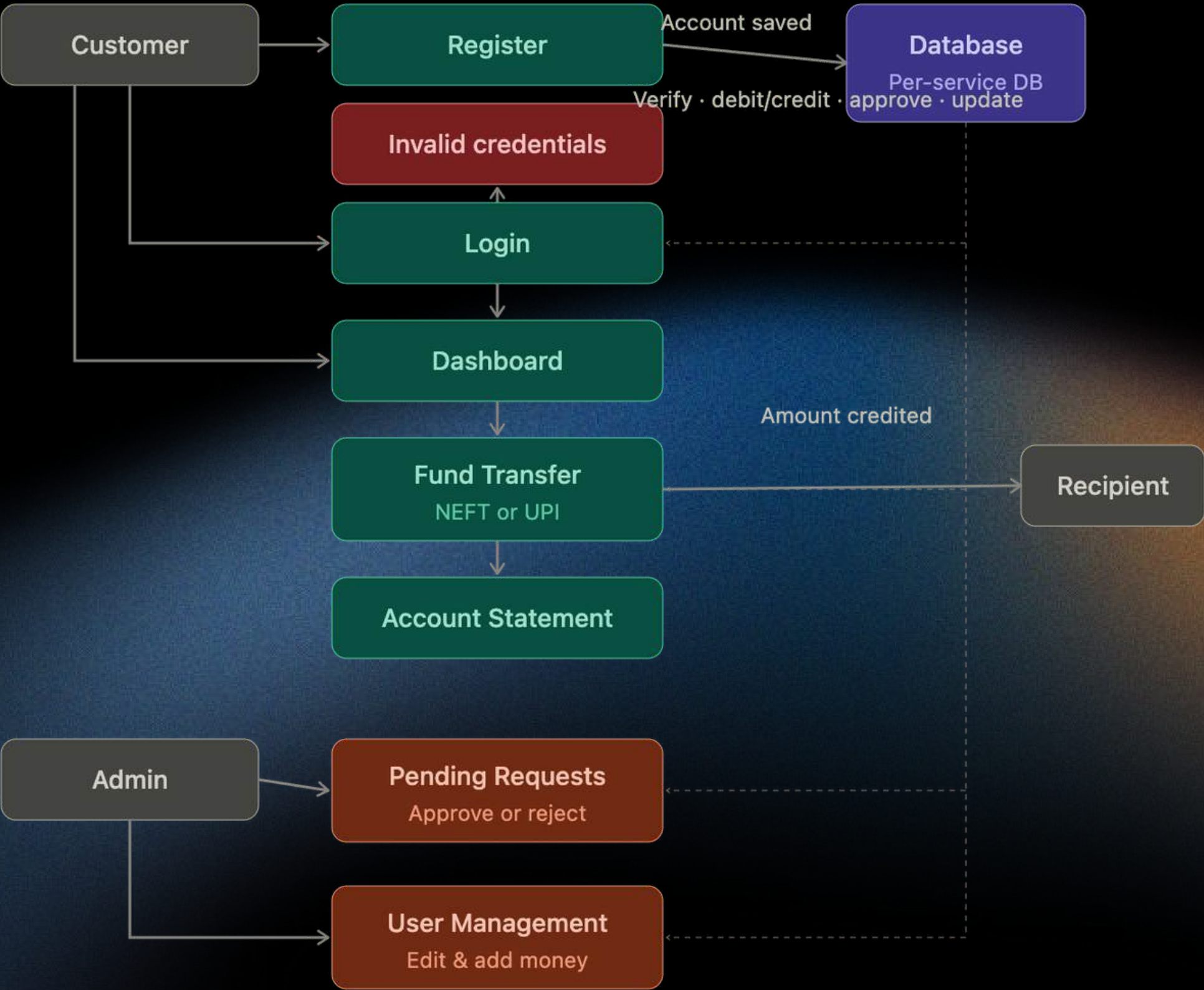
PROJECT MODULES

- **Authentication Module (Auth Service)** – Registration, login, OTP, JWT issuing, password management.
- **Account Management Module (Account Service)** – Account opening requests, admin approval, profile, dashboard.
- **Transaction Module (Transaction Service)** – NEFT/UPI fund transfers, payee management, transaction statements.
- **Admin Module** – Account approvals, user management, fund credit, system-wide transaction monitoring.
- **API Gateway** – Single entry point, request routing, JWT validation.
- **Service Registry (Eureka)** – Dynamic service discovery across all microservices.
- **Frontend Module (Angular)** – Unified user interface for customers and admins with glassmorphism design.

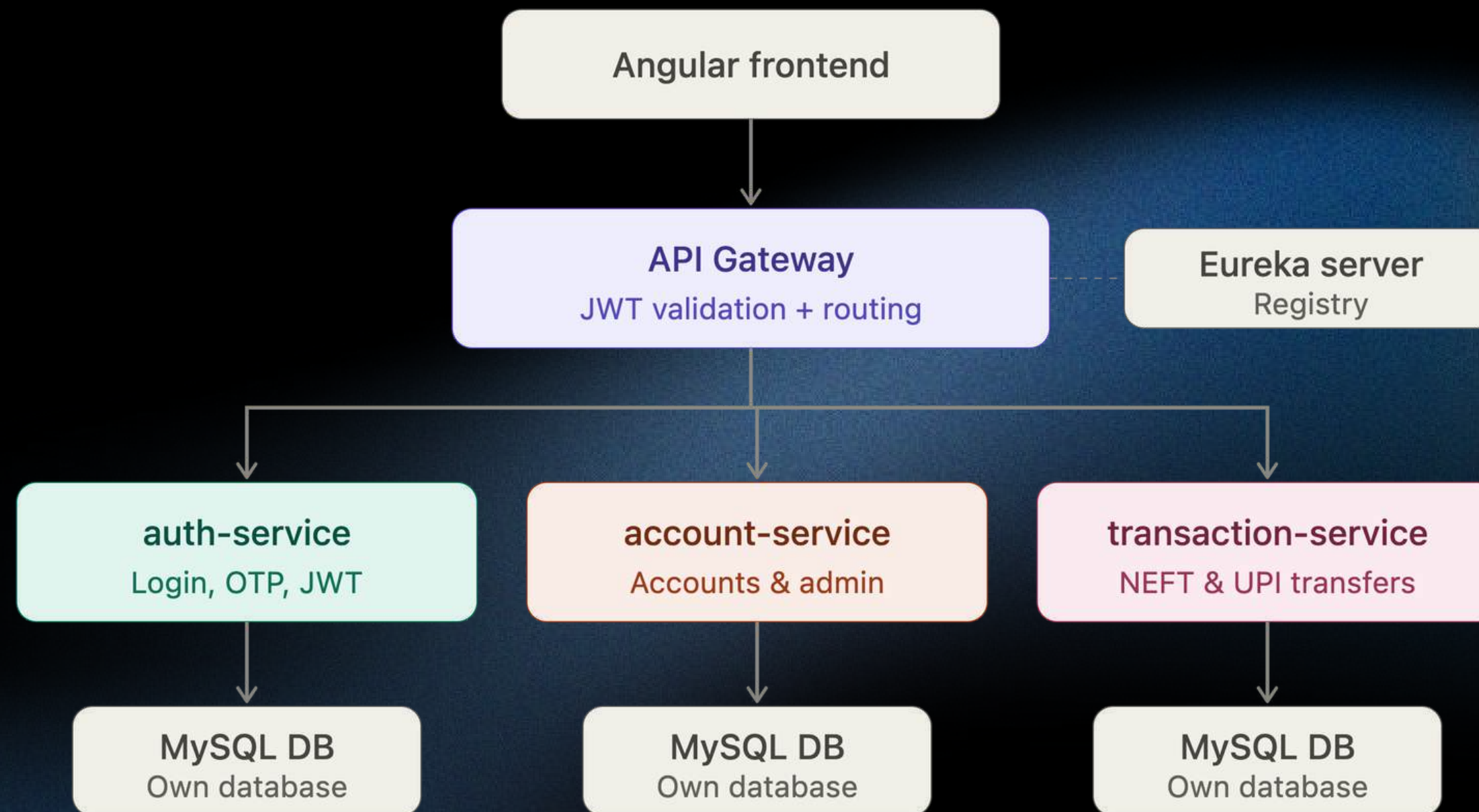
USER CHARACTERISTICS

- **Customers** – General public users with basic computer/smartphone literacy; require a simple, intuitive interface for banking tasks.
- **Administrators** – Bank staff responsible for verifying account requests, managing users, and monitoring transactions; require access to a centralized control panel.
- **System expects users to have:** basic internet familiarity, a valid mobile number/email for OTP verification, and access to a modern web browser.
- **No technical expertise required** for customers — the system is designed to be self-explanatory and beginner-friendly.

SEQUENCE DIAGRAM

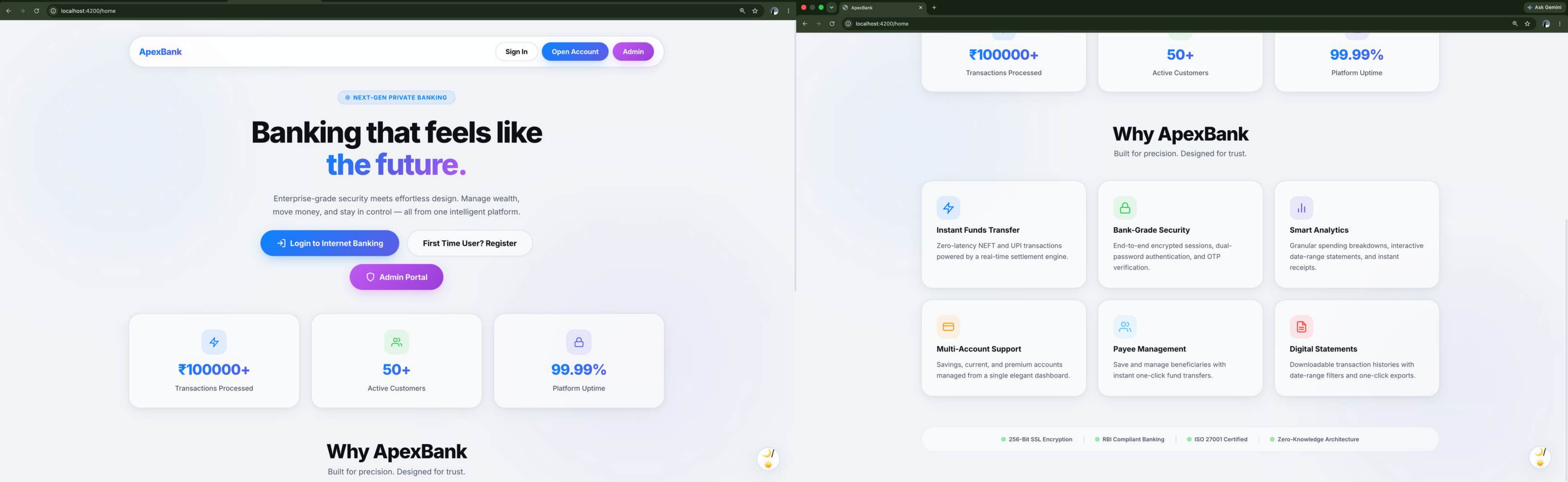


PROJECT ARCHITECTURE

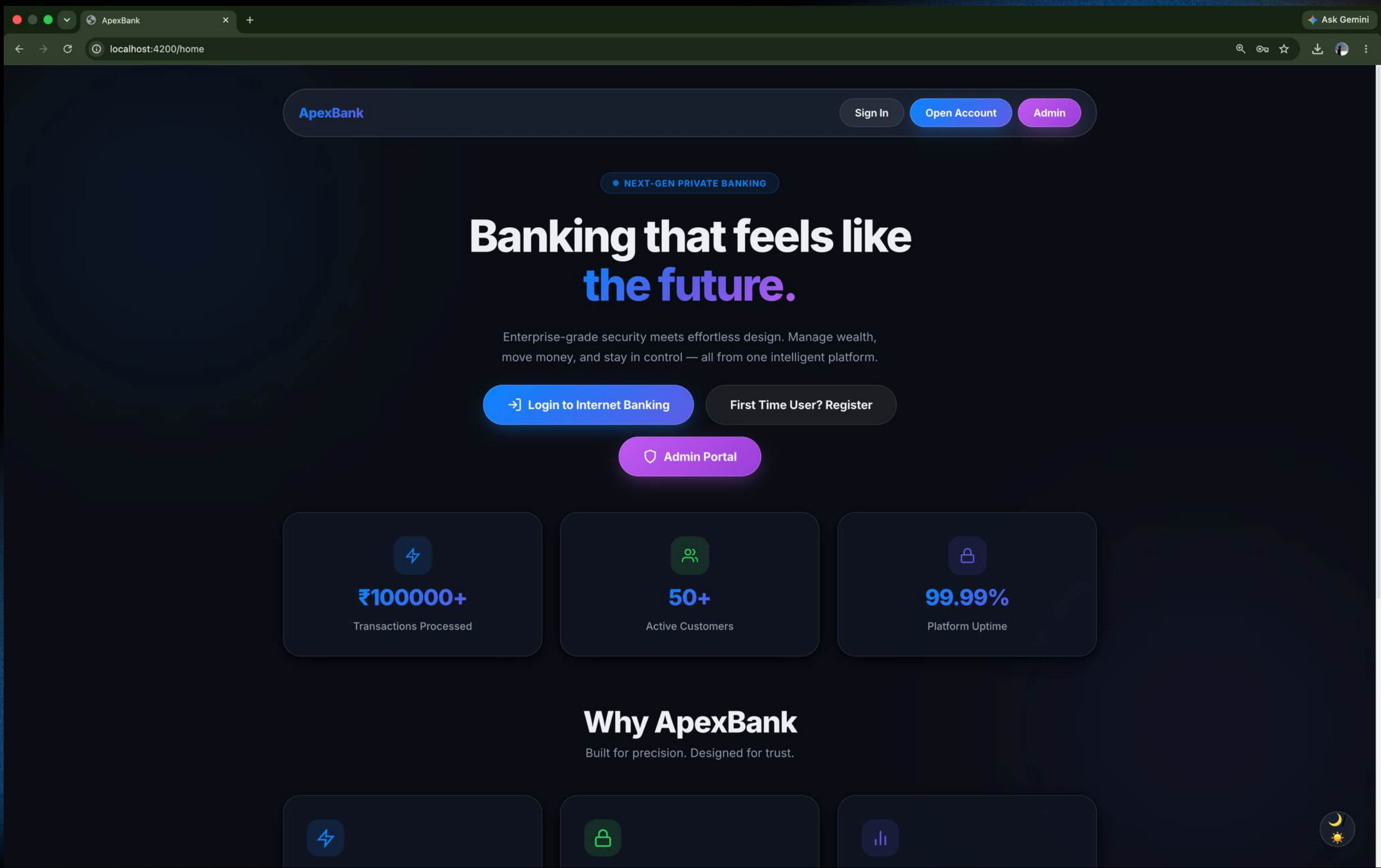


PROJECT SCREENSHOTS

HOME PAGE



DARK THEME



OPEN A NEW SAVINGS ACCOUNT

ApexBank

Open a Savings Account

Complete the form below to apply for a new ApexBank account

Personal Details

TITLE

Mr

FIRST NAME

First name

MIDDLE NAME

Middle name (optional)

LAST NAME

Last name

FATHER'S NAME

Father's full name

MOBILE NUMBER

10-digit mobile number

EMAIL ID

email@example.com

AADHAR CARD NUMBER

12-digit Aadhar number

DATE OF BIRTH

dd/mm/yyyy



Residential Address

ADDRESS LINE 1

House/Flat No., Building Name

ADDRESS LINE 2

Street, Area, Locality

ADDRESS LINE 2

Street, Area, Locality

LANDMARK

Nearby landmark

STATE

State

CITY

City

PINCODE

6-digit pincode

Occupation Details

OCCUPATION TYPE

Select occupation

SOURCE OF INCOME

e.g. Salary, Business

GROSS ANNUAL INCOME

Select income range

Debit Card & Net Banking



I want a Debit/ATM Card



Opt for Net Banking



I agree to the Terms & Conditions

Submit Application

[← Back to Home](#)

CREATING INTERNET BANKING

ApexBank

Ask Gemini

localhost:4200/register

ApexBank

Create Internet Banking Access

Set up your online banking credentials

ACCOUNT NUMBER

Enter your account number

LOGIN PASSWORD

Minimum 8 charact

CONFIRM LOGIN PASSWORD

Re-enter login pass

TRANSACTION PASSWORD

Minimum 6 charact

CONFIRM TRANSACTION PASSWORD

Re-enter transactio

OTP VERIFICATION

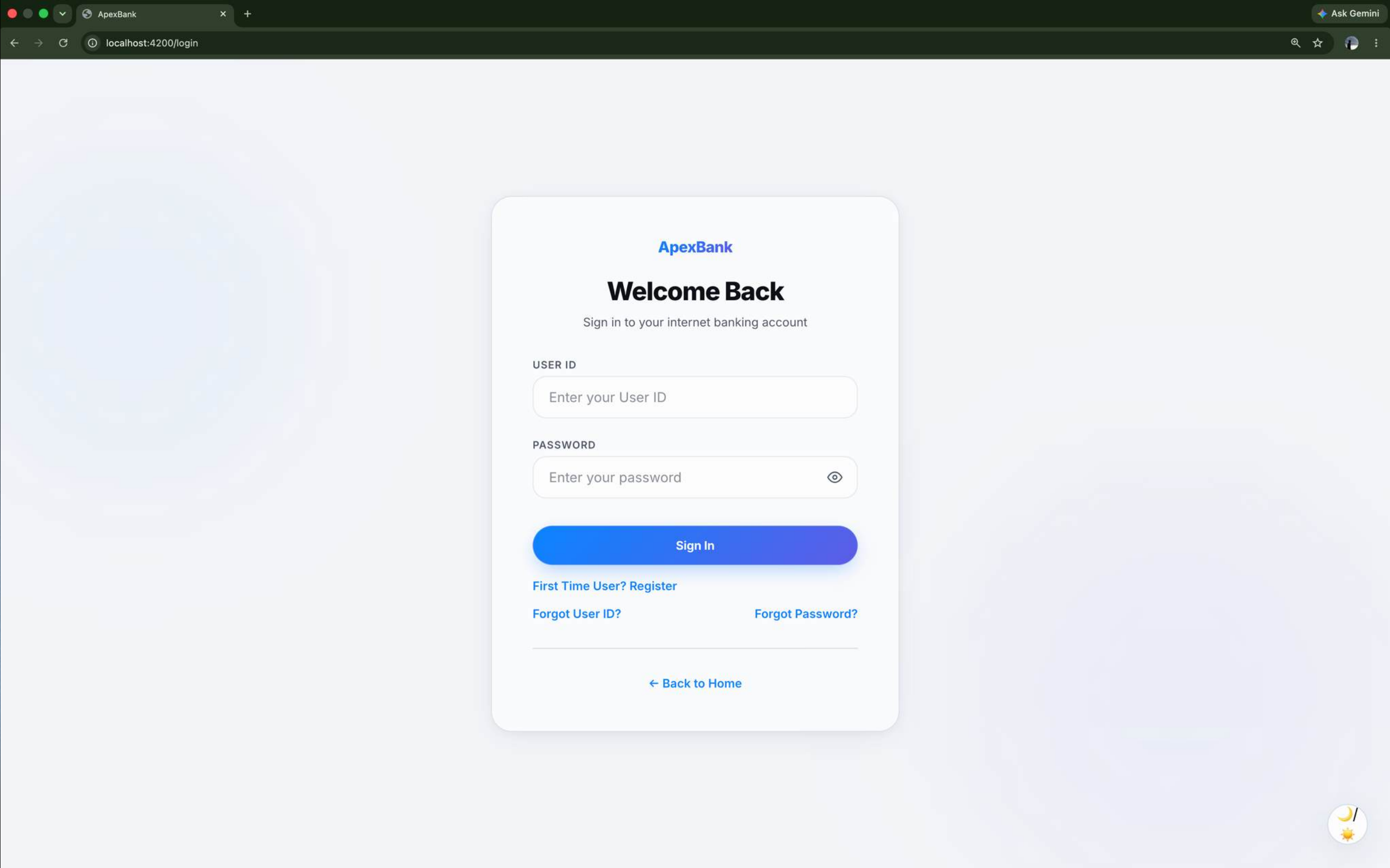
6-digit OTP

Send OTP

Activate Internet Banking

[← Already registered? Login](#)

SIGN IN / LOGIN PAGE



USER DASHBOARD

A

ApexBank

PRIVATE BANKING

S

Srivatsav Reddy Na...

A/C: 147150885397

OVERVIEW

ACCOUNTS

STATEMENTS

TRANSFER MONEY

BENEFICIARIES

USER PROFILE

SECURITY

Sign Out

Good day, Srivatsav 🙌

Here's your financial snapshot for today.

TOTAL BALANCE

₹ 199,908.00

A/C 5397

SAVINGS

YOUR UPI ID

8019962379@apex

Send Money

Transfer

Add Payee

Statement

Summary

Profile

Recent Transactions

↓

To 130094803330

UPI · Aug 5, 2026 · 11:19 PM

-₹5,678.00

↑

From 130094803330

NEFT · Aug 5, 2026 · 11:17 PM

+₹3,920.00

View all →

🌙

🌞

ACCOUNTS PAGE

A

ApexBank

PRIVATE BANKING

S

Srivatsav Reddy Na...

A/C: 147150885397

OVERVIEW

ACCOUNTS

STATEMENTS

TRANSFER MONEY

BENEFICIARIES

USER PROFILE

SECURITY

Sign Out

Account Summary

A comprehensive breakdown of your account and digital card access.

ApexBank

WORLD ELITE

..... 5397

ACCOUNT HOLDER
147150885397

VISA

Account Details

ACTIVE

Account Number

147150885397

Available Balance

₹ 199,908.00

Currency

INR (₹)

Status

OPERATIONAL

Recent Transactions

View Full Statement →

REFERENCE ID	MODE	TO / FROM A/C	AMOUNT	DATE & TIME
REF9521746923615	UPI	130094803330	-₹5,678.00	Aug 5, 2026 · 11:19 PM
REF9520638201902	NEFT	130094803330	+₹3,920.00	Aug 5, 2026 · 11:17 PM
REF8415080831536	NEFT	130094803330	-₹12,334.00	Aug 4, 2026 · 4:35 PM
REF8412463221369	UPI	130094803330	-₹800.00	Aug 4, 2026 · 4:30 PM
REF8405982217121	NEFT	130094803330	-₹100.00	Aug 4, 2026 · 4:19 PM

ACCOUNTS STATEMENT PAGE

A

ApexBank

PRIVATE BANKING

S

Srivatsav Reddy Na...

A/C: 147150885397

OVERVIEW

ACCOUNTS

STATEMENTS

TRANSFER MONEY

BENEFICIARIES

USER PROFILE

SECURITY

Sign Out

Account Statement

Filter, view, and export your account transactions by date range.

FROM DATE

01/08/2026

TO DATE

06/08/2026

Generate Statement

Print Statement

TOTAL CREDIT INFLOW

+₹11,920.00

TOTAL DEBIT OUTFLOW

-₹42,012.00

TOTAL TRANSACTIONS

9

REFERENCE ID	MODE	DIRECTION	TO / FROM A/C	AMOUNT	REMARKS	DATE & TIME
REF8326254967215	NEFT	DEBIT	130094803330	-₹500.00	Coffee	Aug 4, 2026 · 2:07 PM
REF8330450949244	NEFT	CREDIT	130094803330	+₹8,000.00	Rent	Aug 4, 2026 · 2:14 PM
REF8331535443287	UPI	DEBIT	130094803330	-₹3,400.00	HouseOfSpirits(split)	Aug 4, 2026 · 2:15 PM
REF8333619489868	NEFT	DEBIT	155067677785	-₹19,200.00	Concert Tickets	Aug 4, 2026 · 2:19 PM
REF8405982217121	NEFT	DEBIT	130094803330	-₹100.00	gift	Aug 4, 2026 · 4:19 PM
REF8412463221369	UPI	DEBIT	130094803330	-₹800.00	Coffee	Aug 4, 2026 · 4:30 PM
REF8415080831536	NEFT	DEBIT	130094803330	-₹12,334.00	gift	Aug 4, 2026 · 4:35 PM

FUND TRANSFER PAGE

ApexBank

PRIVATE BANKING

S

Srivatsav Reddy Na...

A/C: 147150885397

OVERVIEW

ACCOUNTS

STATEMENTS

TRANSFER MONEY

BENEFICIARIES

USER PROFILE

SECURITY

Sign Out

Fund Transfer

Instant, zero-latency transfers via NEFT & UPI.

NEFT Transfer

Instant UPI

Pay via UPI ID

Pay via Account No.

BENEFICIARY UPI ID

8019962376@apex

TRANSFER AMOUNT (₹)

₹ 400

REMARKS / NOTE (OPTIONAL)

Stationary

TRANSACTION PASSWORD

.....

Confirm UPI Payment

Saved Beneficiaries

+ Add New

D

D***

A/C: 130094803330

>

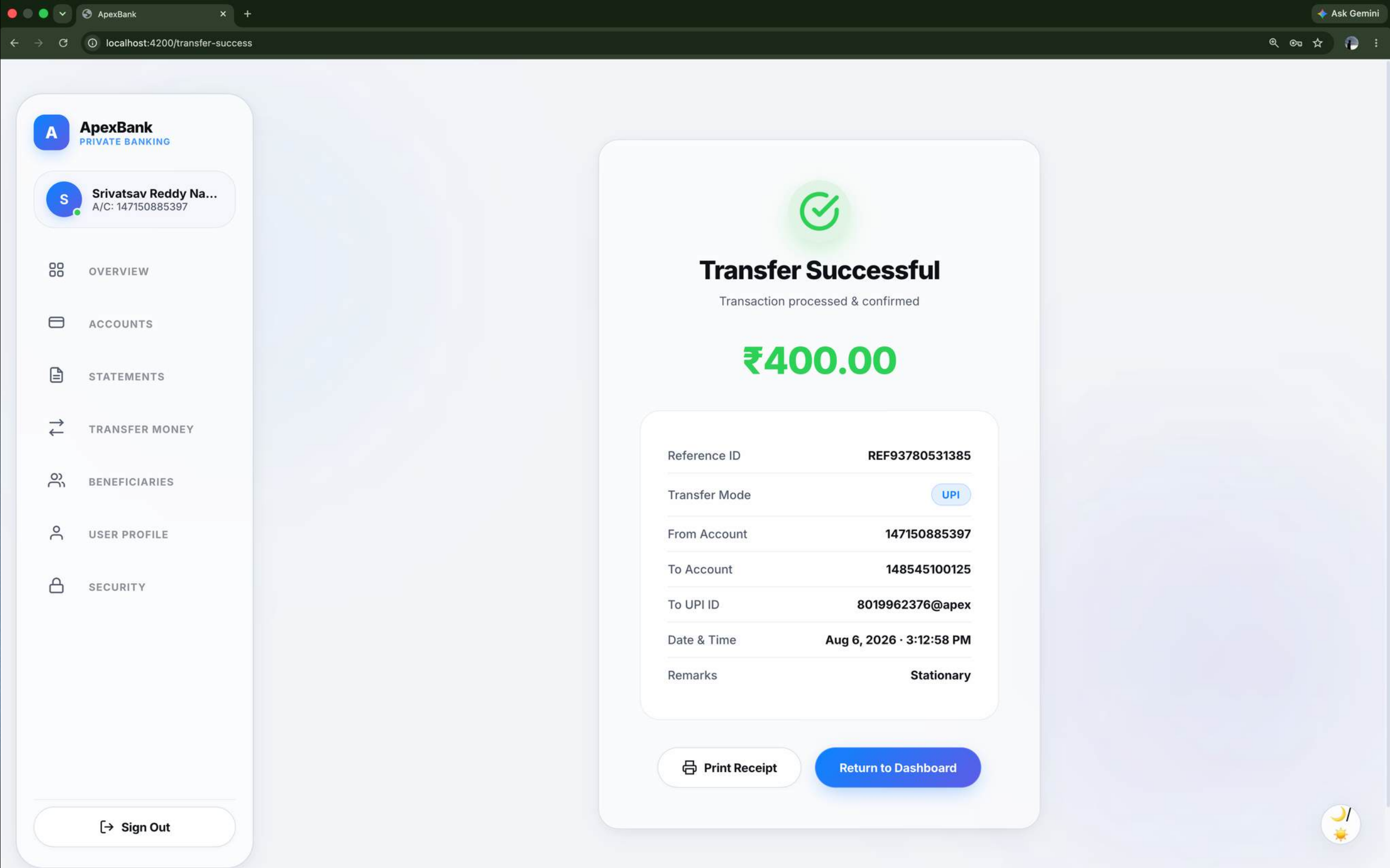
S

Sushank

A/C: 155067677785

>

TRANSACTION SUCCESS PAGE



ADD BENEFICIARY PAGE

A

ApexBank

PRIVATE BANKING

S

Srivatsav Reddy Na...

A/C: 147150885397

OVERVIEW

ACCOUNTS

STATEMENTS

TRANSFER MONEY

BENEFICIARIES

USER PROFILE

SECURITY

→ Sign Out

Add New Beneficiary

Save a payee for quick fund transfers.

BENEFICIARY NAME

BENEFICIARY ACCOUNT NUMBER

RE-ENTER ACCOUNT NUMBER

NICKNAME (OPTIONAL)

Save Beneficiary

Saved Beneficiaries

D

D***
130094803330

S

Sushank
155067677785

Go to Fund Transfer →

ApexBank

x

+

<

>

↺

🔒 localhost:4200/dashboard/profile

🔗

☆

👤

⋮

A

ApexBank
PRIVATE BANKING

S

Srivatsav Reddy Na...
A/C: 147150885397

☰ OVERVIEW

📁 ACCOUNTS

📄 STATEMENTS

↕ TRANSFER MONEY

👥 BENEFICIARIES

👤 USER PROFILE

🔒 SECURITY

➡ Sign Out

User Profile

Manage your personal identification, contact information, and address records.

SN

Mr Srivatsav Reddy Nagam
A/C: 147150885397 UPI: 8019962379@apex

Edit Profile

Contact & Identification

MOBILE NUMBER

8019962379

EMAIL ADDRESS

apexuser1@gmail.com

Verified Identity Badges

AADHAR NUMBER

567856785678

DATE OF BIRTH

2005-06-15

Residential Address

ADDRESS LINE 1

Cbit,Hyderabad

ADDRESS LINE 2

Cbit

LANDMARK

cbit

STATE

Telangana

CITY

Hyderabad

PINCODE

500075

🌙

🌞

CHANGE PASSWORD PAGE

ApexBank

localhost:4200/dashboard/change-password

Ask Gemini

A

ApexBank

PRIVATE BANKING

S

Srivatsav Reddy Na...

A/C: 147150885397

OVERVIEW

ACCOUNTS

STATEMENTS

TRANSFER MONEY

BENEFICIARIES

USER PROFILE

SECURITY

[-> Sign Out]

Change Password

Update your login and transaction passwords securely.

Login Password

CURRENT LOGIN PASSWORD

Enter current login password

NEW LOGIN PASSWORD

Minimum 8 characters

CONFIRM NEW LOGIN PASSWORD

Re-enter new login password

Transaction Password

CURRENT TRANSACTION PASSWORD

Enter current 6-digit transaction password

NEW TRANSACTION PASSWORD

Minimum 6 characters

CONFIRM NEW TRANSACTION PASSWORD

Re-enter new transaction password

Update Passwords

ADMIN LOGIN PAGE

ApexBank

Ask Gemini

localhost:4200/admin/login

ApexBank



Admin Portal

Restricted access — authorised personnel only

ADMIN ID

PASSWORD



 Sign In as Admin

[← Back to Home](#)



ADMIN DASHBOARD PAGE

A

ApexBank

ADMIN PORTAL

A

Admin User

ADMIN

Overview

Account Requests

All Users

Transactions

Credit Account

→

Sign Out

Admin Dashboard

System overview and key metrics



23

Total Users



₹39,82,109.10

Total Balance



0

Pending Requests



0

Total Transactions



ACCOUNT REQUESTS PAGE

ApexBank

ADMIN PORTAL

Admin User

ADMIN

Overview

Account Requests

All Users

Transactions

Credit Account

➔ Sign Out

Account Requests

Review and process account opening applications

Pending Approvals

All Requests

RS

Mr Rah...

9876543210

APPROVED

Salaried · Applied Jan 10, 2024

Review Application →

PR

Ms Pri...

9876543211

APPROVED

Salaried · Applied Feb 5, 2024

Review Application →

AP

Mr Ami...

9876543212

APPROVED

Business · Applied Feb 28, 2024

Review Application →

SI

Ms Sn...

9876543213

APPROVED

Salaried · Applied Mar 15, 2024

Review Application →

VS

Mr Vik...

9876543214

APPROVED

Salaried · Applied Apr 8, 2024

Review Application →

AN

Ms An...

9876543215

APPROVED

Salaried · Applied Apr 25, 2024

Review Application →

RM

Mr Roh...

9876543216

APPROVED

Business · Applied May 12, 2024

Review Application →

KR

Ms Ka...

9876543217

APPROVED

Salaried · Applied May 28, 2024

Review Application →

AG

Mr Arj...

9876543218

APPROVED

Salaried · Applied Jun 15, 2024

Review Application →

DK

Ms Div...

9876543219

APPROVED

Student · Applied Jun 30, 2024

Review Application →

SK

Mr Sur...

9876543220

APPROVED

Salaried · Applied Jul 15, 2024

Review Application →

MD

Ms Me...

9876543221

APPROVED

Salaried · Applied Aug 1, 2024

Review Application →

KS

Mr Kar...

9876543222

APPROVED

Salaried · Applied Aug 15, 2024

Review Application →

PV

Ms Po...

9876543223

APPROVED

Salaried · Applied Aug 20, 2024

Review Application →

AI

Mr Adi...

9876543224

APPROVED

Salaried · Applied Aug 25, 2024

Review Application →

SI

Mr Sus...

9876543225

APPROVED

Salaried · Applied Aug 30, 2024

Review Application →

🌙

☀️

USER DETAILS PAGE

Account Requests

All Users

Transactions

Credit Account

→ Sign Out

VS

Mr Vikram...

100000000005

ACTIVE

Balance: ₹5,89,740.00 · SAVINGS

View Details →

AN

Ms Anany...

100000000006

ACTIVE

Balance: ₹5,89,740.00 · SAVINGS

View Details →

RM

Mr Rohan ...

100000000007

ACTIVE

Balance: ₹5,89,740.00 · SAVINGS

View Details →

KR

Ms Kavith...

100000000008

ACTIVE

Balance: ₹93,420.80 · SAVINGS

View Details →

AG

Mr Arjun ...

100000000009

ACTIVE

Balance: ₹4,12,670.15 · SAVINGS

View Details →

MD

Ms Meera ...

100000000012

ACTIVE

Balance: ₹1,98,760.25 · SAVINGS

View Details →

KS

Mr Karthi...

100000000013

ACTIVE

Balance: ₹4,56,890.00 · CURRENT

View Details →

AM

Mr Arjun ...

100000000016

ACTIVE

Balance: ₹50,000.00 · SAVINGS

View Details →

SI

Mr Susha...

100722771102

ACTIVE

Balance: ₹480.00 · SAVINGS

View Details →

SI

Mr Susha ...

155087877785

ACTIVE

Balance: ₹1,24,520.00 · SAVINGS

View Details →

SN

Mr Srivats...

147150885397

ACTIVE

Balance: ₹1,99,908.00 · SAVINGS

View Details →

DA

Mr Dikshit...

119131783143

ACTIVE

Balance: ₹1,13,004.00 · SAVINGS

View Details →

Mr James Bond

ACTIVE

Account Number

148545100125

Account Type

SAVINGS

Email

james@gmail.com

Occupation

Self Employed

Address

Cbit,Hyderabad , Hyderabad, Telanagana - 456789

Created

Aug 5, 2026

Balance

₹20,000.00

Mobile Number

8019962376

Date of Birth

Jun 12, 1986

Debit Card

Yes

Last Updated

Aug 5, 2026

Close Window

NEW USERS AMOUNT CREDIT PAGE

A

ApexBank

ADMIN PORTAL

A

Admin User

ADMIN

Overview

Account Requests

All Users

Transactions

Credit Account

→ Sign Out

Credit Account

Add money to any user's account

ACCOUNT NUMBER

Enter account number

AMOUNT (₹)

Enter amount to credit

REMARKS (OPTIONAL)

Enter remarks for this credit...

₹ Credit Account

Ask Gemini

TRANSACTIONS TRACKING PAGE

A

ApexBank

ADMIN PORTAL

A

Admin User

ADMIN

Overview

Account Requests

All Users

Transactions

Credit Account

↪ Sign Out

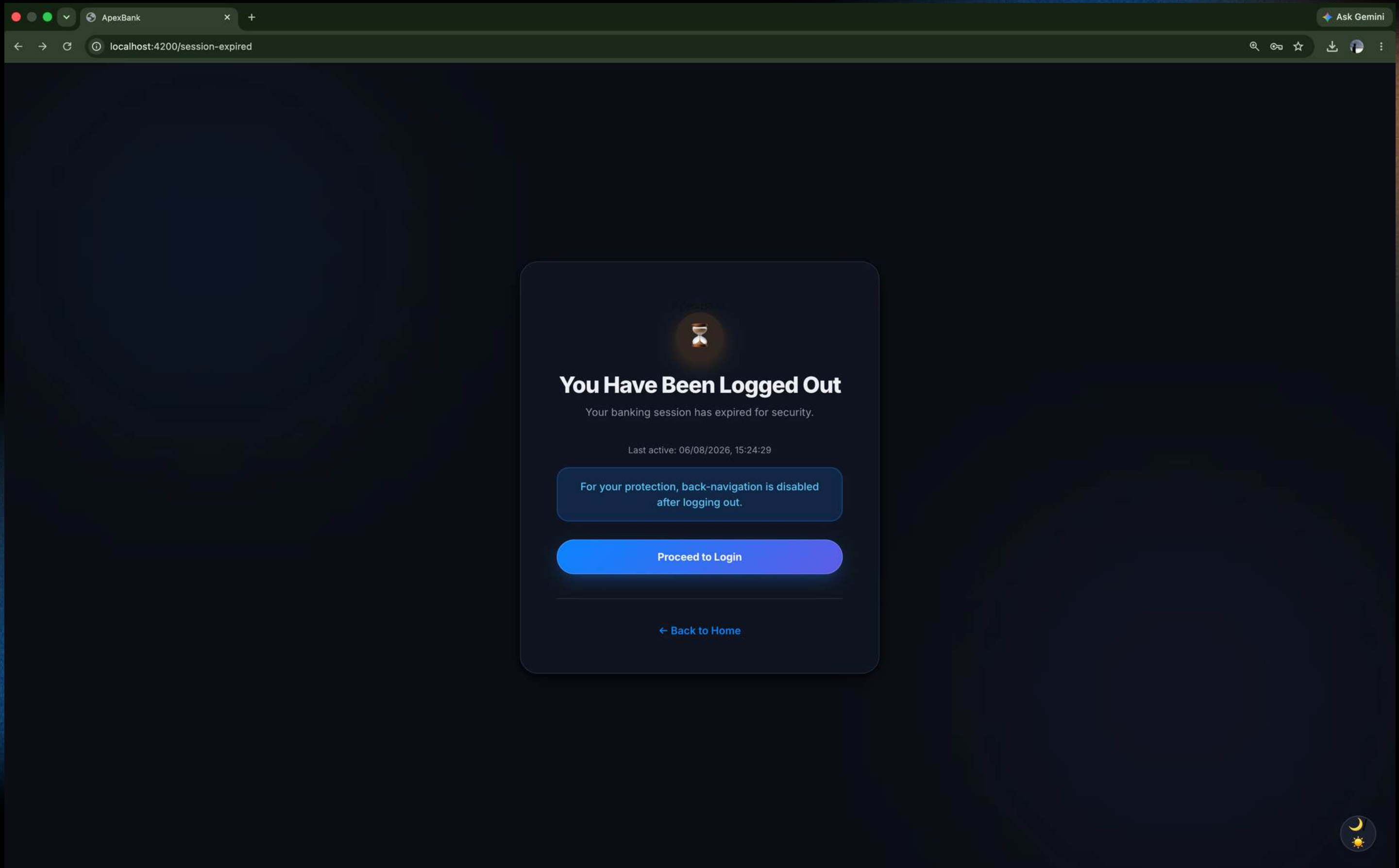
All Transactions

System-wide transaction history

REFERENCE ID	MODE	FROM ACCOUNT	TO ACCOUNT	AMOUNT	DATE
REF9521746923615	UPI	147150885397	130094803330	₹5,678.00	Aug 5, 2026 23:19
REF9520974748765	UPI	130094803330	155067677785	₹300.00	Aug 5, 2026 23:18
REF9520638201902	NEFT	130094803330	147150885397	₹3,920.00	Aug 5, 2026 23:17
REF9520070438909	NEFT	130094803330	181905490098	₹500.00	Aug 5, 2026 23:16
REF8415080831536	NEFT	147150885397	130094803330	₹12,334.00	Aug 4, 2026 16:35
REF8412463221369	UPI	147150885397	130094803330	₹800.00	Aug 4, 2026 16:30
REF8405982217121	NEFT	147150885397	130094803330	₹100.00	Aug 4, 2026 16:19
REF8333619489868	NEFT	147150885397	155067677785	₹19,200.00	Aug 4, 2026 14:19
REF8331535443287	UPI	147150885397	130094803330	₹3,400.00	Aug 4, 2026 14:15
REF8330450949244	NEFT	130094803330	147150885397	₹8,000.00	Aug 4, 2026 14:14
REF8330051633788	UPI	130094803330	155067677785	₹5,500.00	Aug 4, 2026 14:13
REF8326254967215	NEFT	147150885397	130094803330	₹500.00	Aug 4, 2026 14:07
REF6630996377667	UPI	100722771102	155067677785	₹20.00	Aug 2, 2026 15:01
REF6628552595679	UPI	155067677785	100722771102	₹500.00	Aug 2, 2026 14:57
TXN202507300006	UPI	100000000004	100000000008	₹1,200.00	Jul 30, 2025 22:10
TXN202507300010	UPI	100000000008	100000000014	₹450.00	Jul 30, 2025 13:50
TXN202507290008	UPI	100000000006	100000000010	₹999.00	Jul 29, 2025 21:25
TXN202507290012	UPI	100000000010	100000000002	₹650.00	Jul 29, 2025 18:30

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LOGOUT PAGE



FUTURE SCOPE

- Integrate real SMS/Email gateways for live OTP delivery, replacing the current simulated OTP system.
- Add mobile applications (Android/iOS) and biometric-based two-factor authentication for enhanced accessibility and security.
- Introduce AI-based fraud detection to monitor transactions and flag suspicious activity in real time.

THANK YOU!